

# THE GEOMETRIC BOUNCE HYPOTHESIS

An APH Derivation of the Lunar Origin via Vacuum Phase Transition

Aaron M. Schutz

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## Abstract

We present a resolution to the Isotopic Crisis of lunar formation within the Axiomatic Physical Homeostasis (APH) framework. We propose that the Moon is not the result of an external impact (Theia), but the product of a **Geometric Bounce** event on the Proto-Earth. We model this event as a critical failure of a planetary-scale Geometric Stiffness Reactor (GSR), leading to a transient excursion of the vacuum stiffness parameter  $\beta \gg 1.91$ . This creates a repulsive *Associator Pressure* capable of overcoming gravitational binding energy, ejecting the silicate mantle while retaining the iron core. This hypothesis naturally derives the identical isotopic composition of the Earth-Moon system and the Moon's anomalous low density as consequences of **Associative Purification**.

## 1 Introduction: The Isotopic Crisis

The standard Giant Impact Hypothesis faces a fine-tuning problem: the Earth and Moon possess virtually identical isotopic signatures for Oxygen ( $^{17}\text{O}/^{16}\text{O}$ ), Titanium ( $^{50}\text{Ti}/^{47}\text{Ti}$ ), and Tungsten ( $^{182}\text{W}$ ). Since solar system bodies exhibit distinct isotopic fingerprints based on their formation distance from the Sun, an external impactor (Theia) should have contaminated the mixing pool. The likelihood of a Theia with an Earth-identical composition is vanishingly small ( $< 1\%$ ).

We propose an endogenic origin: the Moon consists entirely of Proto-Earth material ejected by an internal repulsive force derived from the non-associative geometry of the vacuum.

## 2 The Physics of the Geometric Bounce

### 2.1 The Stiffened Vacuum Equation of State

In the APH framework, the vacuum possesses a Geometric Stiffness  $\beta$ . Under standard conditions,  $\beta \approx 1$  (Linear/Associative). In high-energy regimes stabilized by APH control mechanisms (such as the GSR), the stiffness becomes super-linear,  $\beta_{QCD} \approx 1.91$ .

The effective pressure  $P_{geom}$  exerted by the vacuum against matter density  $\rho$  is given by the modified polytropic equation of state:

$$P_{geom}(\rho) = K\rho^\Gamma \quad \text{where} \quad \Gamma = 1 + \beta \quad (1)$$

For a stable reactor core,  $\Gamma \approx 2.91$ . This supports the core against gravity. However, a control failure leading to an *Associator Hazard Excursion* creates a runaway stiffness parameter  $\beta(t) \rightarrow \infty$ .

## 2.2 The Bounce Condition

Consider the Proto-Earth with mass  $M_{\oplus}$  and radius  $R$ . The gravitational pressure scales as  $P_{grav} \propto R^{-4}$ . The geometric repulsive pressure scales as  $P_{geom} \propto R^{-3\Gamma}$ . The ejection condition is defined when the geometric repulsion exceeds the gravitational binding energy:

$$\int P_{geom} dV > \frac{3GM_{\oplus}^2}{5R} \quad (2)$$

With  $\Gamma \approx 2.91$ , the repulsive force scales as  $F_{rep} \propto R^{-6.73}$ . A rapid spike in  $\beta$  (the  $\beta$ -Excursion) creates a shockwave where  $P_{geom} \gg P_{grav}$  at the core-mantle boundary. This results in the explosive ejection of the overlying crust and mantle.

## 3 Associative Purification: Derivation of Lunar Composition

### 3.1 The Iron Depletion Mechanism

The Moon is anomalously depleted in iron (core mass  $\approx 1 - 3\%$  of total) compared to Earth ( $\approx 33\%$ ). APH explains this via **Associator Sorting**.

The GSR mechanism functions by concentrating the Associator Hazard  $\mathcal{A}(Z)$  into the reactor core to induce stiffness. In APH, mass is proportional to the Associator Hazard. Therefore, the heaviest elements (Iron/Nickel) act as the *fuel* or anchors for the stiffness reaction and are retained in the deep potential well of the reactor (Earth's Core).

The ejected material is the **Associative Residue**—matter with lower Associator Hazard (Silicates/Oxides) that is expelled by the stiffened vacuum pressure.

$$\rho_{Moon} \approx \rho_{Earth} - \langle \mathcal{A}(Z)_{Core} \rangle \quad (3)$$

This predicts the Moon is composed of *Geometric Slag*—purified associative matter—matching the observed density of  $\approx 3.3 \text{ g/cm}^3$  (Mantle density) versus Earth's  $\approx 5.5 \text{ g/cm}^3$ .

### 3.2 Isotopic Identity

Since the Moon is formed from the Proto-Earth's own mantle ejected via vacuum repulsion, the isotopic signatures  $\delta^{17}O$ ,  $\delta^{50}Ti$ , etc., are necessarily identical by definition. No mixing of an external reservoir is required.

## 4 Quantitative Predictions

### 4.1 The Separation Energy

The energy required to eject the Moon mass  $M_L \approx 0.012M_{\oplus}$  to the Roche limit is  $E_{eject} \approx 3 \times 10^{30} \text{ J}$ . In the APH framework, this energy is derived from the relaxation of the vacuum stiffness following the excursion. The Latent Heat of the Phase Transition ( $\Delta\beta$ ) is:

$$E_{vac} \approx V_{core} \cdot \Lambda_{GUT} \cdot (\beta_{peak} - \beta_{stable}) \quad (4)$$

This implies the event was a release of stored vacuum potential energy, not chemical or kinetic impact energy.

### 4.2 Core-Mantle Boundary Anomalies

The APH model predicts that the Earth's Core-Mantle Boundary (CMB) should exhibit scars from this event. Specifically, we predict the existence of **Large Low-Shear-Velocity Provinces (LLSVPs)** not as thermal plumes, but as regions of **Residual Geometric Stiffness** ( $\beta > 1$ ) left over from the reactor operation. These regions should exhibit anomalous seismic wave propagation speeds consistent with a non-associative bulk modulus correction.

## 5 Conclusion

The existence of the Moon is the macroscopic physical evidence of a planetary-scale vacuum phase transition. The APH formulation of the Geometric Bounce resolves the isotopic crisis, explains the iron depletion, and provides a unified mechanism for the thermal history of the Earth-Moon system. We conclude that the Moon is the cooled remnant of the Earth's associative mantle, ejected by a failure of homeostatic control in the pre-biotic era.